

```
#include <Geotopia>
#include <Kernel File>
#include <string>
int main() {
    std::ifstream inputFile("vocabulary.txt"); // Open a geometry file named
vocabulary.txt
```

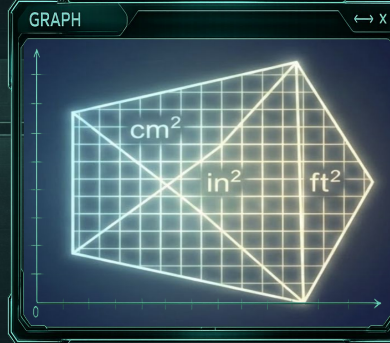
Internet Security of Geotopia



Access Granted

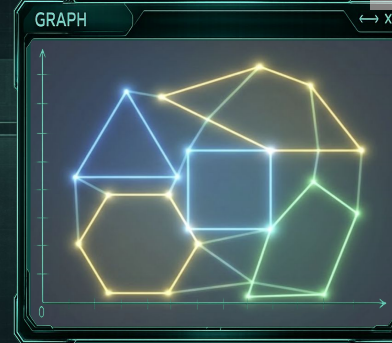
DEFINITION

Area is the amount of space inside a 2D shape. Measured in square units (cm^2 , in^2 , ft^2).



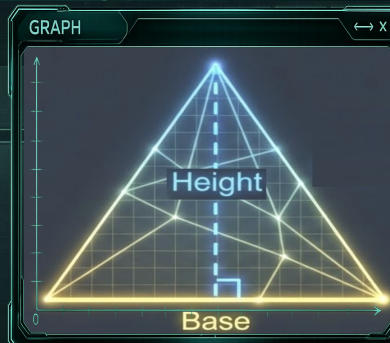
DEFINITION

Polygon is a closed flat shape made of straight line segments.



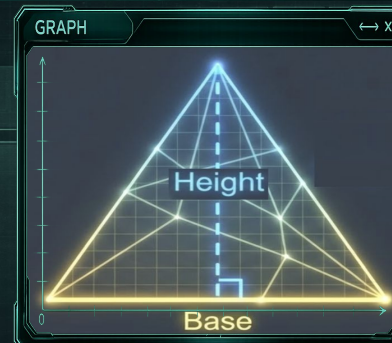
DEFINITION

Base (of a triangle) is the side you choose to measure height from.



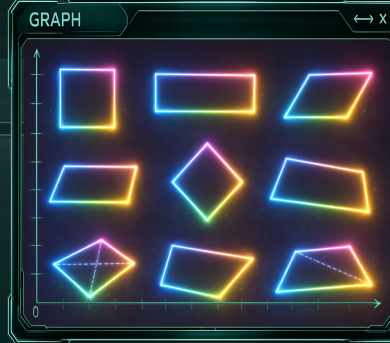
DEFINITION

Height is the distance straight up and down from the base to the vertex (top corner) or opposite side.



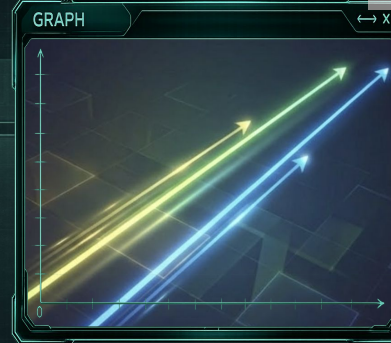
DEFINITION

Quadrilateral is a polygon with 4 sides.



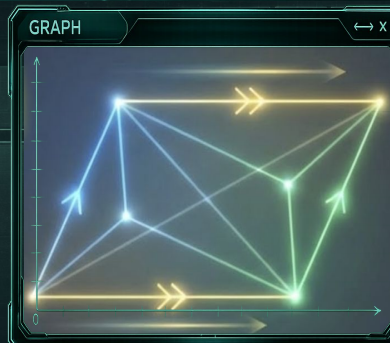
DEFINITION

Parallel lines go in the same direction and never meet, even if they keep going forever.



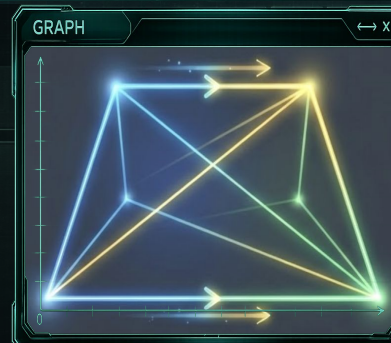
DEFINITION

Parallelogram is a quadrilateral where opposite sides are parallel. To find its area, multiply base by height.



DEFINITION

Trapezoid is a quadrilateral with one pair of opposite sides that are parallel. To find the area, multiply height by half of the two bases.



DEFINITION

An isosceles triangle has 2 equal sides and angles. To find its area, take half of base times height.

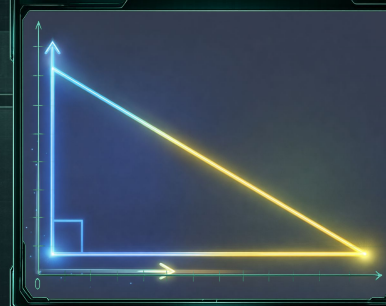
GRAPH



DEFINITION

A right triangle is a triangle that has one 90-degree angle (a square corner). To find its area, take half of base times height.

GRAPH



DEFINITION

An equilateral triangle has 3 equal sides and 3 equal angles of 60° .

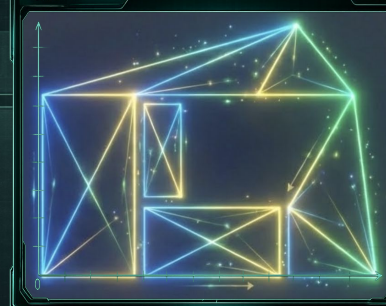
GRAPH



DEFINITION

A composite figure is a polygon made up of basic shapes put together.

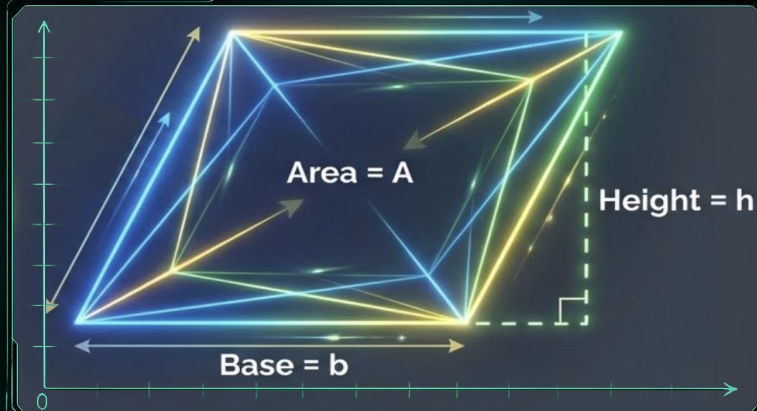
GRAPH



DEFINITION

Find missing length of a parallelogram: Divide the area by the corresponding base or height.

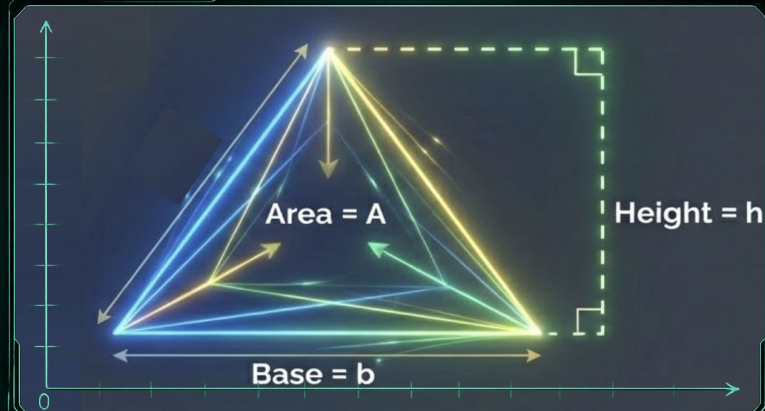
GRAPH



DEFINITION

Find the missing length of a triangle: multiply the area by 2, then divide by either base or height.

GRAPH

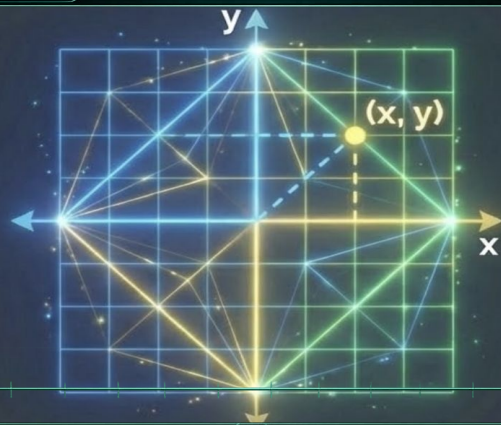


DEFINITION

Coordinate Plane is a graph formed by two number lines that cross at 0.

Ordered Pair: two numbers written as (x, y) that show a point's location.

GRAPH



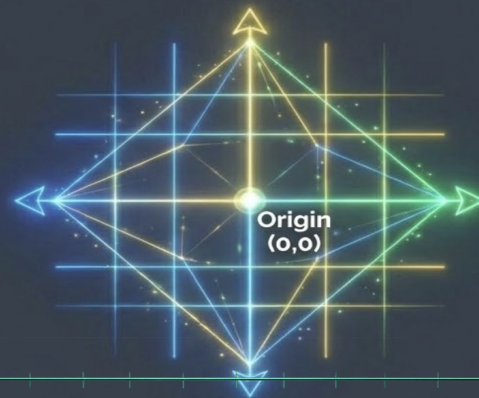
DEFINITION

Origin is $(0,0)$ where x-axis and y-axis meet.

x-axis: the horizontal number line.

y-axis: the vertical number line.

GRAPH

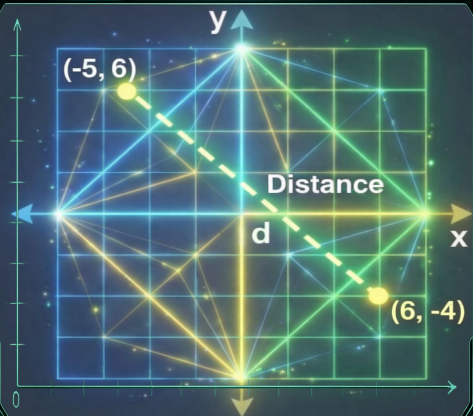


DEFINITION

Distance: how far apart two points are.

Distance between two coordinates: the distance of moving from one point to the other.

GRAPH

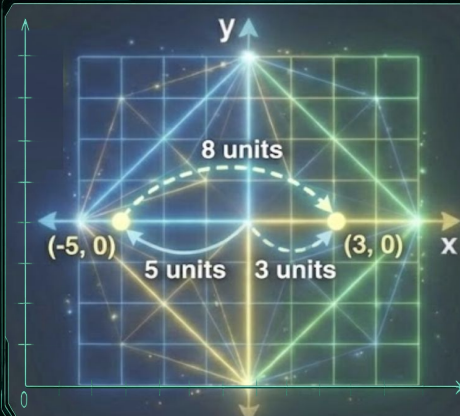


DEFINITION

Example: from $(3, 10)$ to $(-5, 10)$

It's 3 units from 3 to 0, then 5 more units from 0 to -5, for a total of 8 units

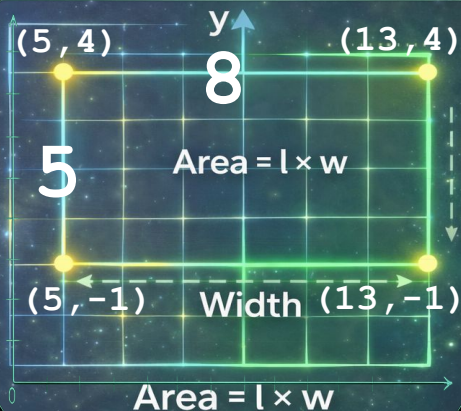
GRAPH



DEFINITION

Find area in a coordinate plane :
using coordinates to calculate
side lengths and applying area
formulas.

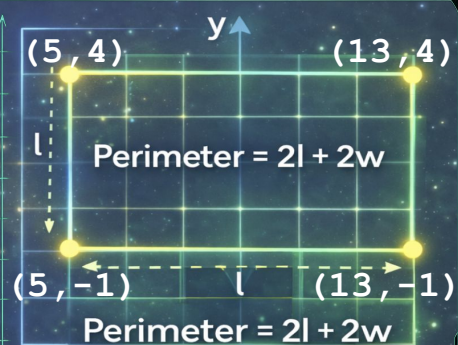
GRAPH



DEFINITION

Find perimeter in a coordinate
plane: using coordinates to
calculate side lengths and add up
for the perimeter.

GRAPH



DEFINITION

Perfect Square: A number that can be made by multiplying a number by itself (forms a square).

First 5 perfect squares: 1, 4, 9, 16, 25

GRAPH



$2 \times 2 = 4$
Square



$6 \times 6 = 36$
Perfect Square

DEFINITION

Perfect Cube: A number that can be made by multiplying a number by itself three times (forms a cube).

First 5 perfect cubes: 1, 8, 27, 64, 125

GRAPH

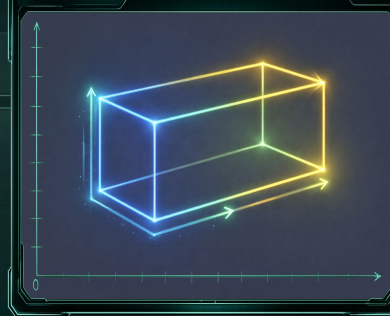


$3 \times 3 \times 3 = 27$
Perfect Cube

DEFINITION

Rectangular prism: a prism with rectangles for all its faces.

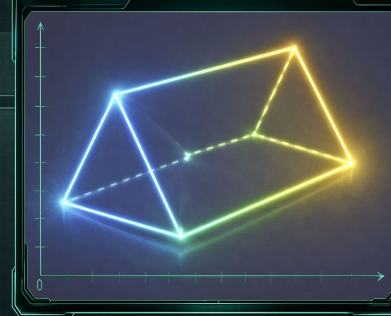
GRAPH



DEFINITION

Triangular prism: a prism with triangular top and bottom faces.

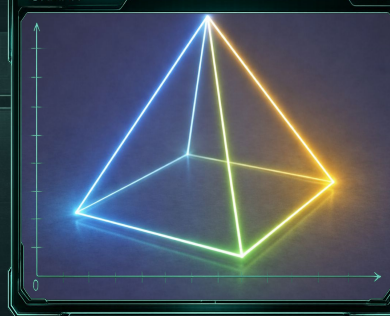
GRAPH



DEFINITION

Square Pyramid: a 3D shape with a square base and 4 faces that meet at one point on top.

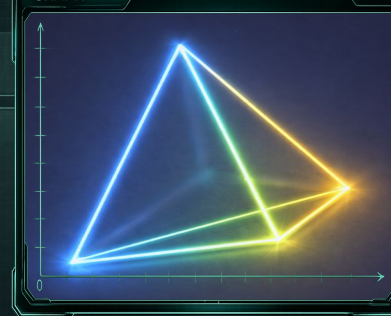
GRAPH



DEFINITION

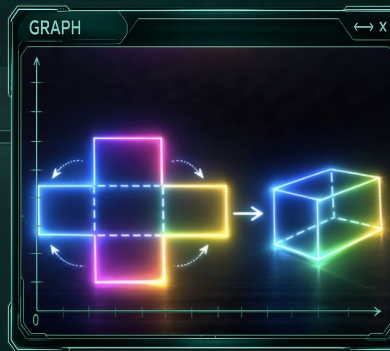
Triangular Pyramid: a 3D shape with a triangle base and 3 faces that meet at one point on top.

GRAPH



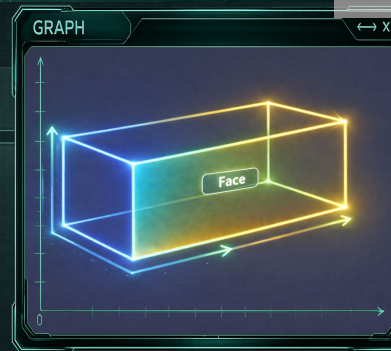
DEFINITION

Net: a pattern that can be folded to make a 3D shape.



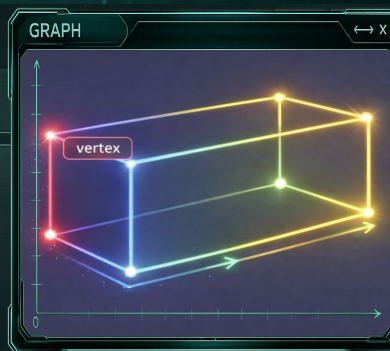
DEFINITION

Face: A flat surface of a 3D shape. It is like one "side" of a solid figure.



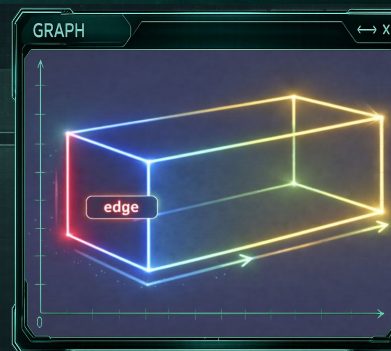
DEFINITION

Vertex (plural: vertices): A corner point where two or more edges meet.



DEFINITION

Edge: A line segment where two faces meet.



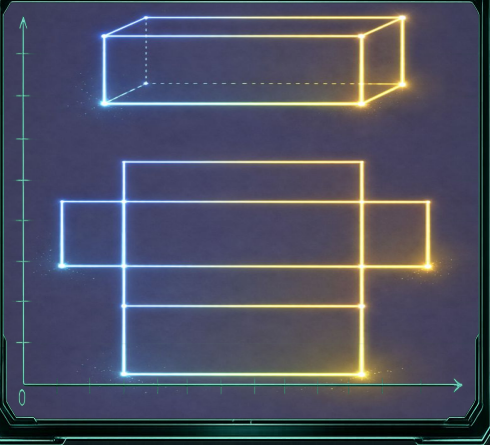
DEFINITION



Surface area: the total area of all the outside faces of a 3D shape.

How to find surface area: find the area of each face, then add the answers up.

GRAPH



DEFINITION

Volume: how much space a 3D shape takes up inside. (in unit such as cm^3 , in^3 , ft^3)

Volume of rectangular prism: multiply its length, width by its height.

GRAPH

